

What Makes a Good Fantasy Football Manager?

Evidence from one league over ten years and 252 leagues over one

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Abstract

Fantasy football managers hold two incompatible beliefs at once: that the game is mostly luck, and that some managers are reliably good. We reconcile them by decomposing weekly team scores into persistent manager quality and week-to-week noise, and deriving the probability that the better manager wins. We began expecting to find a low and general ceiling on skill. That is not what the data show. In one fourteen-team league observed over four seasons the intraclass correlation is 0.039 — a manager one standard deviation above average wins only 55.7% of his weeks, and 0.110 estimated season by season — but across 252 public leagues the median is 0.294, where the same manager wins 67.6%. Skill content is therefore a property of the *league*, not of the game. About half the difference is attributable to disengagement rather than judgment: dropping the two lowest-scoring managers from each league moves the median from 0.294 to 0.164, and the typical worst team scores 80% of its league average against 93% in the tightly-contested league we study in detail. Within that league, what most distinguishes managers is early-round draft outcomes ($r = 0.805$ with persistent scoring) — but draft performance does not persist across eras ($r = -0.157$), so what looks like drafting skill is largely a draft that worked. The one behaviour that is both learnable and demonstrably rewarded is waiver claim volume.

1 Introduction

Ask a fantasy football league who its best manager is and you will get a confident answer. Ask the same people whether the game is mostly luck and most will say yes. Both convictions are usually held by the same person, and neither is usually accompanied by a number.

They are not, in fact, contradictory. If persistent differences in manager quality are small relative to week-to-week scoring noise, a real and stable advantage still produces a win rate near a coin flip in any single week, and a fourteen-game season is far too short to reveal it. Whether that describes a given league is an empirical question with a specific answer, and the answer turns out to vary enormously from league to league.

We set out to estimate the ceiling on manager skill, expecting to find it low and roughly universal. The first half of that was right for the league we know best and wrong in general. Our contributions are:

1. A variance decomposition of weekly scores into persistent manager quality and noise, and a closed-form map from it to the probability that the better manager wins (Section 3).
2. Detailed estimates for one fourteen-team league over four seasons, where $\rho = 0.039$ and the attainable edge is very small (Section 4.1).
3. The same estimate in 252 public leagues, where the median is 0.294 and the studied league sits near the 15th percentile on a like-for-like basis — falsifying the hypothesis that the ceiling is general (Section 4.2).

4. Evidence that roughly half of the typical league’s apparent skill gap is disengagement rather than judgment (Section 4.3).
5. A within-league analysis of which manager behaviours track quality, and the finding that the strongest one does not persist across a decade (Sections 4.5–4.6).

Related work

Becker and Sun [2016] formulate draft and lineup management as a mixed-integer program, with weekly scoring modelled bilinearly as the product of a player’s innate ability at a statistic and his opponent’s ability to defend it. Two of their results anchor ours. Their Table 3 reports mean absolute weekly forecast errors of 6.77 points at running back against a mean of 7.46, and 4.61 at tight end against 4.72: the typical miss is 87–98% of the typical score. And when their optimizer is granted perfect knowledge of the season’s outcomes, it wins a ten-team league 24.7% of the time against a 10% baseline. Omniscience is worth a factor of 2.5.

Lutz [2015] applies support vector regression and neural networks to six seasons of quarterback data, reaching a best mean absolute error of 6.22 points and a mean relative error of 0.42, without materially beating the linear baselines he compares against.

Both papers ask how well weekly scores can be *predicted*. We ask what the irreducible residual implies for the attainable edge, and we work backwards from realised outcomes rather than forwards from projections. We are not aware of a published variance budget for head-to-head fantasy football, nor of an estimate of how much that budget varies across leagues.

2 Data

The focal league. A fourteen-team, half-PPR, head-to-head league running since 2016, with a starting lineup of QB / RB / RB / WR / WR / TE / FLEX / WRRB_FLEX / K / DEF and four bench slots. It has used three platforms: MyFantasyLeague 2016–19, Yahoo 2020–21, Sleeper since 2022. Primary analysis uses the Sleeper era, from which we obtain every draft pick, every weekly matchup with the full starter list and per-player points, every roster, and every transaction *including failed waiver claims with their bid amounts*. This gives 770 manager-weeks, 798 draft picks, and 1,089 reconstructable waiver auctions. We restrict to weeks 1–14, the fantasy regular season.

The earlier eras enter only in Section 4.6. Of those, only the 2016–17 drafts are recoverable: 2018 returns picks with an empty player field, 2019 returns no draft results, and the 2020–21 Yahoo export contains standings and matchups but no player-level data. The MyFantasyLeague seasons are also structurally different — twenty franchises drafting as two *independent* ten-team drafts from a shared player pool — and stored no scoring rules, so player values there are computed in half-PPR and may not match what that league actually scored.

The comparison sample. To ask whether the focal league is typical we snowball-sampled public Sleeper leagues, alternating between collecting the leagues a known user belongs to and the members of a newly seen league, starting from the focal league’s members. We retained completed 2025 seasons with at least eight managers and ten scored weeks, yielding **252 leagues**. Only aggregate quantities are kept: league size, scoring type, roster positions, and the matrix of weekly scores with managers reduced to integers. No usernames, team names, or league names from third-party leagues are stored or reported.

This is a convenience sample reached through one league’s social graph, and is not a random draw from all Sleeper leagues. Section 6 returns to what that does and does not threaten.

3 Methods

3.1 A season as an accounting identity

Before any model, note that a season decomposes exactly. Let O_m be the points manager m ’s *optimal* lineup would have scored and S_m what he actually started. Every started player was either drafted by him or acquired later, so

$$O_m = \underbrace{D_m}_{\text{drafted starters}} + \underbrace{A_m}_{\text{acquired starters}} + \underbrace{L_m}_{\text{left on the bench}}, \quad (1)$$

with no residual. This is arithmetic, not estimation, and it is where we begin because it fixes the scale of what follows.

3.2 Variance components

Let Y_{mt} be the points scored by manager m ’s starting lineup in week t . We posit

$$Y_{mt} = \mu + \alpha_m + \varepsilon_{mt}, \quad \alpha_m \sim (0, \tau^2), \quad \varepsilon_{mt} \sim (0, \sigma^2), \quad (2)$$

with $\alpha_m \perp \varepsilon_{mt}$, and define the intraclass correlation $\rho = \tau^2 / (\tau^2 + \sigma^2)$.

We stress what α_m is not. It is everything that persists week to week — including a draft that broke well, and including a manager who has stopped setting his lineup. It is an upper bound on skill, not a measurement of it. Much of this paper is about the gap between the two.

3.3 From variance to win probability

A matchup is decided by the margin $D_{mm't} = (\alpha_m - \alpha_{m'}) + (\varepsilon_{mt} - \varepsilon_{m't})$. The talent gap enters once; the noise enters *twice*, with variance $2\sigma^2$. Under approximate normality,

$$\Pr(m \text{ beats } m' \mid \delta) = \Phi\left(\frac{\delta}{\sigma\sqrt{2}}\right), \quad \delta = \alpha_m - \alpha_{m'}. \quad (3)$$

Writing $\delta = c\tau$ for a manager c standard deviations above average and defining the **skill-to-noise ratio**

$$\kappa \equiv \frac{\tau}{\sigma\sqrt{2}} = \sqrt{\frac{\rho}{2(1-\rho)}}, \quad (4)$$

this collapses to

$$\boxed{\Pr(\text{win a given week}) = \Phi(c\kappa)} \quad (5)$$

The second equality in (4) follows from $\tau^2 = \rho(\tau^2 + \sigma^2)$ and $\sigma^2 = (1 - \rho)(\tau^2 + \sigma^2)$, and is the reason everything downstream — win rates, season records, the observation horizon — follows from ρ alone. With n independent matchups, $W \sim \text{Bin}(n, \Phi(c\kappa))$.

3.4 Estimation

Panels are unbalanced, so (2) is fitted by unbalanced one-way ANOVA [Searle et al., 1992]: $\hat{\sigma}^2 = \text{MS}_W$ and $\hat{\tau}^2 = \max\{(\text{MS}_B - \text{MS}_W)/n_0, 0\}$ with $n_0 = (N - \sum_i n_i^2/N)/(k - 1)$. The correction by n_0 is not cosmetic: in the focal league the naive between-group variance of manager means is 26.2, of which $\hat{\sigma}^2/\bar{n} = 9.5$ is sampling noise in the means themselves. Omitting it overstates the persistent component by a third.

Intervals come from a nonparametric bootstrap resampling *managers*, since observations cluster within manager; $B = 2000$.

4 Results

4.1 One league, in detail

Applying (1) to the focal league gives a first orientation:

	mean per season	between-manager SD
D points from drafted starters	1084	207
A points from acquired starters	418	161
L points left on the bench	121	41
O optimal total	1623	

Two-thirds of a manager’s ceiling comes from his draft, a quarter from the wire, and 7% is forfeited to lineup mistakes.

Table 1 gives the variance decomposition.

	actual lineup		optimal lineup	
	est.	95% CI	est.	95% CI
σ^2 (within manager)	499.5	[450, 550]	520.0	[480, 561]
τ^2 (between managers)	20.3	[1.2, 36.9]	20.6	[0.0, 41.7]
ρ (ICC)	0.039	[0.003, 0.065]	0.038	[0.000, 0.071]
$\Phi(\kappa)$	0.557	[0.514, 0.574]	0.556	[0.500, 0.578]

Table 1: Focal league, $k = 14$ managers, 770 manager-weeks, 2022–2025. The optimal-lineup columns replace each manager’s actual starters with the best legal lineup available from his roster that week.

In natural units $\hat{\sigma} = 22.4$ points against $\hat{\tau} = 4.5$: **about 96% of the variation in a weekly score is within-manager noise.** The matchup margin carries a noise SD of 31.6 points against a talent gap SD of 6.4.

The optimal-lineup columns deserve emphasis. Substituting every manager’s *best possible* lineup for his actual one moves ρ from 0.039 to 0.038. Managers do differ in lineup efficiency — the observed range is 91.1% to 94.1% of optimal — but relative to σ the differences do not register. Perfect hindsight start/sit decisions for everyone would not change who wins this league.

Applying (5), a manager one standard deviation above average wins 55.7% of weeks, expects 7.79 wins in fourteen, and finishes above .500 with probability 0.566. By Spearman–Brown, a single season’s record has reliability 0.362; reaching 0.80 requires about 99 weeks, or seven seasons.

4.2 The ceiling is not general

We expected these figures to be roughly typical. They are not.

	median	IQR	10th–90th pct
ρ (ICC)	0.294	[0.171, 0.381]	[0.080, 0.456]
$\Phi(\kappa)$, $P(+1 \text{ SD wins})$	0.676	[0.626, 0.710]	[0.582, 0.741]

Table 2: 252 public Sleeper leagues, 2025 season. Mean $\sigma = 27.5$ points per week; mean $\tau = 17.6$.

The median league has an ICC of 0.294 and a competent manager there wins 67.6% of his weeks rather than 55.7%. Only 14% of leagues fall below $\rho = 0.10$.

The comparison requires care, because the two estimates are not computed over the same span. Every league in the sample is observed for a single season, whereas Table 1 pools four. Pooling treats a manager as one entity across four entirely different rosters, which attenuates τ^2 ; estimating the focal league season by season instead gives 0.141, 0.029, 0.302 and 0.078 for 2022–25, with a median of 0.110. The like-for-like figure is therefore 0.110, and **the focal league sits near the 15th percentile of the sample** — at the 10th if 2025 alone is used. Unusual, but less extreme than the pooled estimate suggests.

That gap is itself informative. Within a single season 11% of weekly variance is between teams; across four seasons only 3.9% is attributable to the *person*. Most of what separates teams within a year is that year’s roster, not a durable property of the manager — the same conclusion Section 4.6 reaches from draft records.

Whatever the ceiling on skill is, it is not a constant of the game: the sample spans $\rho = 0.080$ to 0.456 between its 10th and 90th percentiles.

4.3 Roughly half the difference is disengagement

Public leagues contain abandoned teams, and an abandoned team scores poorly every week — which is exactly what α_m is built to capture, without being skill in any useful sense. To test this we recomputed each league’s ICC after removing its lowest-scoring managers.

sample	n leagues	median ρ
all managers	252	0.294
drop lowest-scoring manager	252	0.217
drop two lowest	252	0.164

Removing two managers per league nearly halves the median ICC. The supporting descriptive is stark: **the median worst team scores 80% of its league average, and the 10th percentile scores 62%**. In the focal league the worst team scores 93% — every manager sits within 7% of the mean.

Two conclusions follow. First, a substantial share of what a naive variance decomposition calls “manager skill” in a public league is somebody who stopped playing. Second, this does not explain the gap away. Applying the identical treatment to the focal league moves it only from 0.039 to 0.035, because it has no disengaged team to remove: the sample median under drop-two is 0.164 against the focal league’s 0.035, a ratio of 4.7. That league is genuinely, unusually compressed, and the compression is not an artifact of who is dropped.

4.4 League design changes the skill content

Scoring format is associated with the ICC in the direction one would predict from the volatility of the underlying statistics: receptions are far more stable week to week than touchdowns, so reception-heavy scoring should shrink σ without much affecting τ .

	n leagues	median ρ
half-PPR	48	0.224
full PPR	201	0.299
10 teams	35	0.197
11–12 teams	193	0.310
13+ teams	20	0.342

Three standard-scoring leagues are omitted from the upper panel as too few to summarise. Correlation between league size and ICC is +0.138. We report these as associations only: the sample is observational, formats are confounded with the kind of league that adopts them, and the size relation runs opposite to the focal league, which is the largest size class and the least noisy. The practical reading is nonetheless that a commissioner has some control over how much skill his league rewards.

4.5 What separates managers within a league

Within the focal league we estimated each manager’s α_m by empirical-Bayes shrinkage and correlated it against six measured behaviours.

behaviour	r with $\hat{\alpha}_m$
early-round draft value	+0.805
early-round bust avoidance	+0.625
lineup efficiency	+0.448
bid discipline	+0.429
waiver efficiency (pts/\$)	+0.429
waiver claim volume	+0.413

With $n = 14$ managers and six tests this is exploratory. It also carries a mechanical component we do not want to oversell: α_m is the persistent part of weekly scoring, and the persistent part of a roster is mostly what was drafted, so the leading correlation is close to arithmetic.

The waiver results are on much firmer ground, resting on 1,089 auctions rather than sixteen picks per manager. Across the fourteen managers, claims attempted per season correlates with waiver production at $r = +0.888$; price paid per successful claim at $r = -0.799$; and claim *win rate* at $r = -0.369$. The negative sign on win rate is informative: a high win rate means bidding only when certain, and certainty is expensive. Restricting to weeks 1–6 so that acquisition date does not confound the comparison, players won for \$0 returned 75.7 points and players won for \$35 or more returned 78.4 — a 3.6% difference for an unbounded price, with points per dollar falling from 29.9 in the \$1–4 band to 1.7 above \$35. Since FAAB expires worthless and bench slots recycle weekly, the marginal cost of an additional small claim is near zero.

We note one correction that changed this section materially. Our first pass credited a claimed player with all points he scored while rostered. Recomputing against weeks he was actually *started* reduced every manager’s waiver production by a median of 65%, and non-uniformly (49% to 78% across managers), which reordered the table.

4.6 Draft performance does not persist

If drafting is a stable trait, it should correlate across time. For the twelve managers present in both eras, the correlation between standardised draft score in 2016–17 and in 2022–25 is $r = -0.157$, 95% CI $[-0.67, +0.46]$ by Fisher’s z . Eight of twelve changed sign; the manager with the best four-year draft record in the current era had the worst in the earlier one.

This estimate is attenuated by measurement error in both halves. With within-era ICC 0.215 for season draft scores, Spearman–Brown gives reliabilities of 0.523 and 0.354 for four- and two-season means, so the disattenuated correlation is $-0.157/\sqrt{0.523 \times 0.354} = -0.365$. Correcting for attenuation moves the estimate *further* from zero in the negative direction; it cannot rescue a claim of persistence. Supporting $r_{\text{true}} = +0.50$ would have required observing $r_{\text{obs}} = +0.215$.

Pooling all six gradeable seasons, the share of variance in season draft scores attributable to the manager falls from 0.215 to 0.027.

We hold this loosely. The eras differ in league size, roster structure and possibly scoring, and four years separate them. It is evidence of absence only in the weak sense: we could not detect persistence with the data available.

4.7 What is outside a manager’s control

Two factors commonly blamed or credited can be measured directly. Draft slot is randomly assigned; across 56 manager-seasons its correlation with season wins is $+0.049$. Schedule is not: regressing wins on points scored gives $R^2 = 0.484$, and adding points *allowed* raises it to 0.719. **Roughly a quarter of the variance in season records is explained purely by whom a manager happened to be scheduled against.**

4.8 Validation, and where it fails

The model is fitted to weekly scores and makes a testable prediction about season records. If win rates are $\text{Bin}(14, p_m)/14$ with $p_m = \Phi(c_m\kappa)$, the talent-induced SD of the win rate is approximately $2\phi(0)\kappa$.

observed SD of season win rate	0.1517
SD implied by a pure coin flip, $n = 14$	0.1336
residual attributable to talent	0.0718
predicted from the weekly model	0.1137

These do not agree. The weekly model predicts a talent spread in season records 1.6 times larger than the records display. The discrepancy is not fatal — $\hat{\tau}^2$ has interval $[1.2, 36.9]$, and 0.0718 corresponds to a τ near its lower end — but the point estimate in Table 1 should be read as an upper bound. Candidate explanations are non-normality in the margin, which heavier tails would produce, and schedule-induced correlation in opponent quality.

5 Discussion

5.1 Repetition, not insight

Table 3 organises the decisions a manager makes by how often he makes them.

A per-decision edge accumulates linearly in the number of decisions, while the noise in *measuring* that edge grows only as $\sqrt{n}$. Repetition thus improves both how much advantage a manager

decision	per season	between-manager SD of season total
trades	~ 1	~ 30
early draft picks	4	207
weekly lineups	14	41
waiver claims (attempted)	32	161

Table 3: Focal league. Waiver claims are attempts including failed bids, median per manager-season (range 7–92); completed claims median 15. Trades are rare and shrinking: nine in 2022, two in 2025.

can bank and our ability to detect that he has it. This organises the paper’s otherwise scattered findings. Waivers produce the cleanest signal in the data because a manager attempts about thirty a season and the sample contains 1,089 auctions. Draft skill does not survive an era change partly because four meaningful picks a year over four years is sixteen noisy observations. Lineup selection never registers because its per-decision effect — about 2.8 points a week — is an order of magnitude below the 31.6 points of matchup noise it competes with.

5.2 Two different questions

“What makes a good fantasy manager” turns out to have two answers that are routinely conflated.

What makes a manager score more this season? Predominantly a draft that worked ($r = 0.805$). That is largely outside his control and does not carry from one era to the next.

What can a manager actually improve? Waiver behaviour: more claims, at lower prices. Its correlation with quality is weaker ($r = 0.413$), but its feedback loop is fast enough to learn from within a single season.

Confusing the first for the second is, we suspect, why so many managers believe they are good at drafting.

6 Limitations

The comparison sample is not random. It was reached by snowball sampling from one league’s social graph, so leagues socially proximate to the focal league are over-represented, and 201 of 252 are full PPR. This threatens the point estimate of the median ICC. It does not threaten the paper’s central claim, which is about *dispersion*: whatever the population median, we observe ICCs from 0.079 to 0.456 within a single sample, and that range is what falsifies a universal ceiling.

Disengagement is inferred, not observed. We identify likely abandoned teams by low scoring, because the crawl retained only weekly scores. Separating “quit” from “bad but trying” requires transaction counts per league, which we did not collect. The claim that roughly half the typical league’s skill gap is disengagement should be read as consistent with the evidence rather than established by it.

α_m is not skill. It is everything persistent. We have no design that decomposes an individual manager’s α_m into judgment and fortune; the cross-era comparison in Section 4.6 is the closest available and it is underpowered.

Normality. Equation (5) assumes an approximately normal margin. Weekly scores are right-skewed sums of skewed player scores, and the validation failure in Section 4.8 is what fat tails would produce.

Free parameters and leakage. The -50 bust threshold and the ± 12 window in the slot-expectation curve were chosen without formal justification; the ordering of managers is stable across thresholds from -100 to 0 but the magnitude is not. The slot curve is fitted on the same picks it scores, inducing mild leakage. Replacement level is computed from season totals although lineups are set weekly, which inflates all points-above-replacement figures.

Single-season ICCs are noisy. Each comparison league contributes one season, so τ there is estimated from roughly fourteen weeks per manager. The focal league’s own per-season estimates range from 0.029 to 0.302 , which is the scale of instability to expect in any individual league’s figure. Conclusions are drawn from the distribution across leagues, not from any single one.

Estimation. Method-of-moments $\hat{\tau}^2$ is truncated at zero, and the bootstrap inherits that truncation — visible in Table 1, where the optimal-lineup interval is pinned at the boundary. A hierarchical Bayesian model with τ on the log scale, partial pooling across leagues, and league covariates entering as coefficients would estimate this properly and is the obvious next step.

7 Conclusion

The question that titles this paper has a smaller answer than most managers believe, and a more variable one than we expected. In a competitive, fully engaged fourteen-team league, a manager one standard deviation above average wins 55.7% of his weeks and would need seven seasons to demonstrate he is not lucky. In the median public league the same manager wins 67.6% — but about half of that larger edge appears to come from playing against people who have stopped trying, and the comparison narrows further once both sides are estimated over the same span.

For a manager wanting to improve, the practical implication is narrow and somewhat deflating. The decision that most determines how good he looks — the first four rounds of his draft — is the one he gets four attempts at per year and cannot demonstrably learn on a human timescale. The decision he can actually improve is the one that feels like administrative work: many small waiver claims rather than few expensive ones. And a season’s standings, in a league where everybody is trying, are close to uninformative about which of the two he did well.

References

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